

MITESH RAJENDRA PATIL

Full-Stack Engineer • DevOps & Cloud • Real-Time Systems

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Professional Summary

Results-driven full-stack engineer in the final year of B.Tech (IT) with demonstrated experience shipping production-grade systems from a **Mediasoup SFU video platform** supporting multi-party WebRTC sessions, to **AI-integrated developer tools** and **cloud-native deployments on AWS**. Comfortable owning the full delivery lifecycle: system design, implementation, containerisation, and cloud deployment. Seeking a high-impact software engineering role where strong fundamentals in **TypeScript, Node.js, React**, and distributed systems deliver measurable business value.

Technical Skills

Languages: TypeScript, JavaScript (ES2023+), C++, Java, SQL

Frontend: React 18, Next.js 14, Tailwind CSS, HTML5, CSS3, WebRTC APIs

Backend: Node.js, Express, Hono, REST APIs, WebSockets, Mediasoup SFU, Bun

Databases & ORM: PostgreSQL, Redis, Supabase, Prisma ORM

DevOps & Cloud: Docker, AWS EC2, GitHub Actions CI/CD, Cloudflare Workers, Vercel

AI / LLM: Gemini API, Vercel AI SDK, Tavily Search API

Tools & OS: Git, Linux, npm, Bun, VS Code

Fundamentals

- Data Structures & Algorithms:
- Object-Oriented Programming:
- Operating Systems:
- Database Management Systems:
- Computer Networks:

Key Projects

EchoRTC Scalable WebRTC Video Conferencing Platform

React · TypeScript · Node.js · Mediasoup SFU · WebSockets · Docker · AWS EC2 · Gemini AI github.com/mitesh1803/echoRtc

- ▶ Architected and shipped a **production SFU-based video platform** using Mediasoup, supporting multi-party audio/video, screen sharing, and real-time chat architecture comparable to Zoom's server-side media routing.
- ▶ Designed the entire **WebSocket signaling layer from scratch**: room lifecycle management, transport negotiation, producer-consumer state machines, and automatic peer reconnection eliminating client-side race conditions.
- ▶ Containerised all services with **Docker** and deployed on **AWS EC2**; configured TURN/STUN infrastructure for reliable NAT traversal across restricted networks.

commit-gen AI-Powered Git Commit CLI Tool

TypeScript · Node.js · Gemini API · CLI · npm

github.com/mitesh1803/commit-gen

- ▶ Built and published a developer productivity CLI tool that analyzes staged Git diffs and automatically generates **conventional commit messages** using Gemini API, reducing repetitive manual commit writing effort.
- ▶ Packaged and distributed as a **standalone npm-executable utility**, enabling seamless installation and usage across development environments.
- ▶ Focused on developer experience by creating a lightweight, terminal-first workflow that integrates directly into existing Git-based development processes.

shrt.ly URL Shortener with Chrome Extension

TypeScript · Node.js · Express · Chrome Extensions API

github.com/mitesh1803/shrt.ly

- Developed a full-stack URL shortening platform with custom slug generation, redirect handling, and a **per-link click analytics dashboard** for monitoring engagement metrics.
- Designed RESTful APIs for URL creation, lookup, and analytics retrieval, enabling scalable and maintainable backend architecture.
- Built a companion **Chrome extension** enabling one-click shortening of the active browser tab, reducing workflow friction and improving usability.

Education

B.Tech, Information Technology

2021 - 2026

Government College of Engineering, Amravati

Achievements & Impact

- Independently architected a **WebRTC SFU** platform typically developed by multi-engineer teams.
- Built **multiple full-stack projects** spanning AI integration, real-time communication, CLI tooling, and browser extensions - all shipped, deployed, and publicly available on GitHub.
- Hands-on production experience across the full cloud stack: **Docker, AWS EC2, Cloudflare Workers, and Vercel** from container to edge.